Data Quality Check with a FAIR Dataset Using Tableau Prep

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Tool Used: Tableau Prep Builder

Course: Data Quality & Management

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1. Dataset Description

- **Title:** Titanic – Machine Learning from Disaster
- **Source / Link:** https://www.kaggle.com/datasets/yasserh/titanic-dataset
- **Contents:** Passenger records from the RMS Titanic (1912), containing 891 rows and 12 columns. Fields include passenger demographics (age, sex), socio-economic class (Pclass), fare paid, cabin number, port of embarkation, and a binary survival outcome.
- **Why Selected:** The dataset is publicly available, widely documented, and fully FAIR-compliant:
    - **Findable** – Hosted on Kaggle with a clear title, metadata, tags, and a persistent URL
    - **Accessible** – Freely downloadable as a CSV file without any login restriction
    - **Interoperable** – Standard CSV format readable by any analytical tool, including Tableau Prep Builder
    - **Reusable** – Well-documented with an open license, widely used in data science education

The dataset also contains multiple realistic data quality issues , including missing values, incorrect data types, irrelevant fields, inconsistent formatting, and invalid values, making it ideal for practicing data cleaning in Tableau Prep.

2. Identified Data Quality Issues

The following issues were identified upon loading the dataset into Tableau Prep Builder:

#	Issue	Field(s) Affected	Quality Dimension
1	177 missing values (19.9% of rows)	Age	Completeness
2	687 missing values (77.1% of rows)	Cabin	Completeness
3	2 missing values	Embarked	Completeness
4	Zero-fare entries (invalid values)	Fare	Validity
5	Categorical data stored as integer type	Survived, Pclass	Validity
6	Fields irrelevant to any meaningful analysis	PassengerId, Ticket, Name	Relevance
7	Abbreviated port codes instead of full names	Embarked (C, Q, S)	Consistency
8	Non-descriptive, abbreviated field names	SibSp, Parch, Pclass	Traceability

These issues span six of the eight data quality dimensions discussed in the course: **completeness, validity, relevance, consistency, traceability**, and **uniqueness** (verified via row-level duplicate inspection in the Profile pane).

3. Tableau Prep Workflow

The cleaning workflow consists of **6 Clean Steps** followed by an Output step, as shown in the flow canvas. All steps use only the Tableau Prep features covered in class.

Flow Overview:

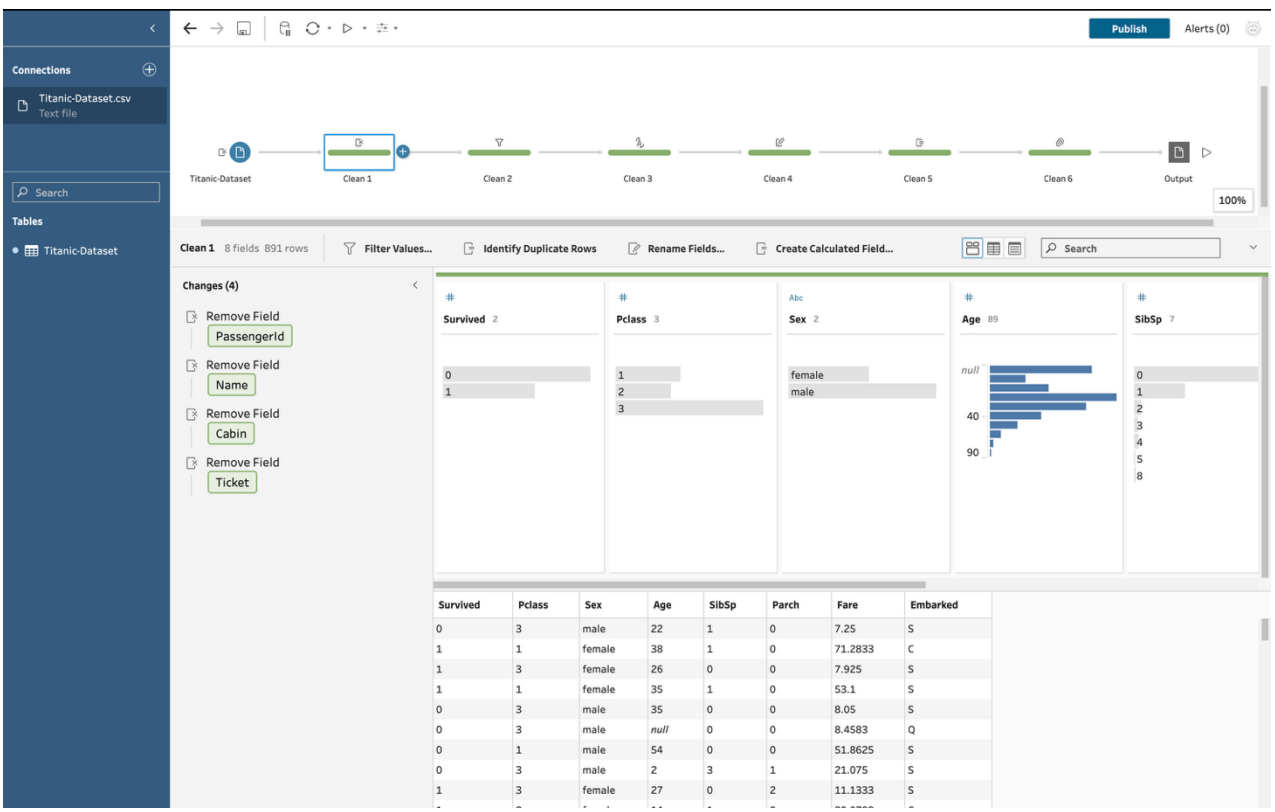
Titanic-Dataset → Clean 1 → Clean 2 → Clean 3 → Clean 4 → Clean 5 → Clean 6 → Output

Step 1 – Input: Load Data

The Titanic-Dataset.csv file is connected via **Connect to Data → Text File**. Tableau Prep auto-detects 12 columns and 891 rows. The Profile pane immediately reveals null values in Age, Cabin, and Embarked, as well as integer types assigned to the categorical fields Survived and Pclass.

Clean 1 – Remove Irrelevant Fields (891 rows → 891 rows, 12 → 8 fields)

Changes (4): Remove Field



The following four fields are removed by right-clicking → **Remove Field**:

- PassengerId – A sequential integer identifier with no analytical value

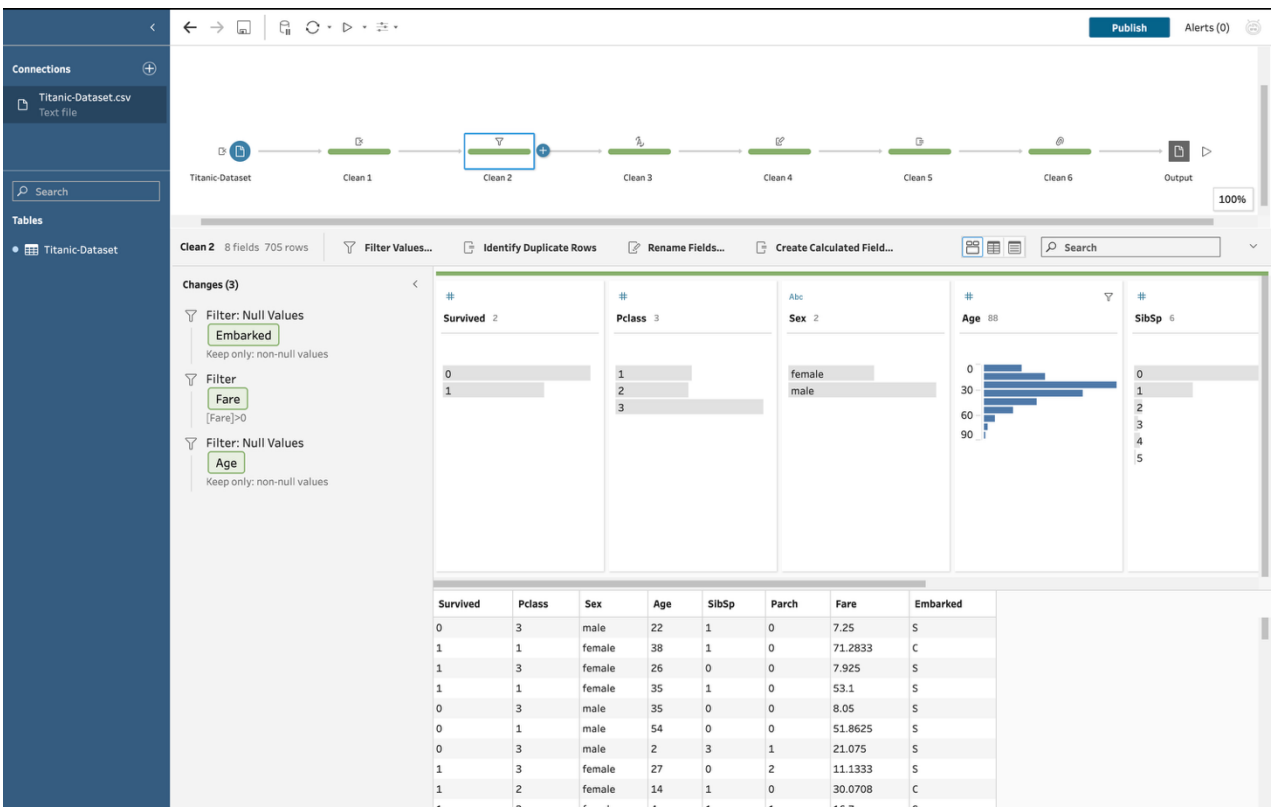
- Name – Free-text passenger names; not useful for structured analysis
- Cabin – Over 77% of values are null, making this field analytically unusable
- Ticket – Alphanumeric ticket codes; inconsistent format and irrelevant to analysis

After this step, 8 fields remain: Survived, Pclass, Sex, Age, SibSp, Parch, Fare, Embarked.

Quality Dimension Improved: Relevance, Completeness

Clean 2 – Filter Invalid and Null Rows (891 rows → 705 rows)

Changes (3): Filter



Three filters are applied to remove incomplete and invalid records:

1. **Filter Null Values – Embarked**: Keep only non-null values → removes 2 rows with missing port of embarkation

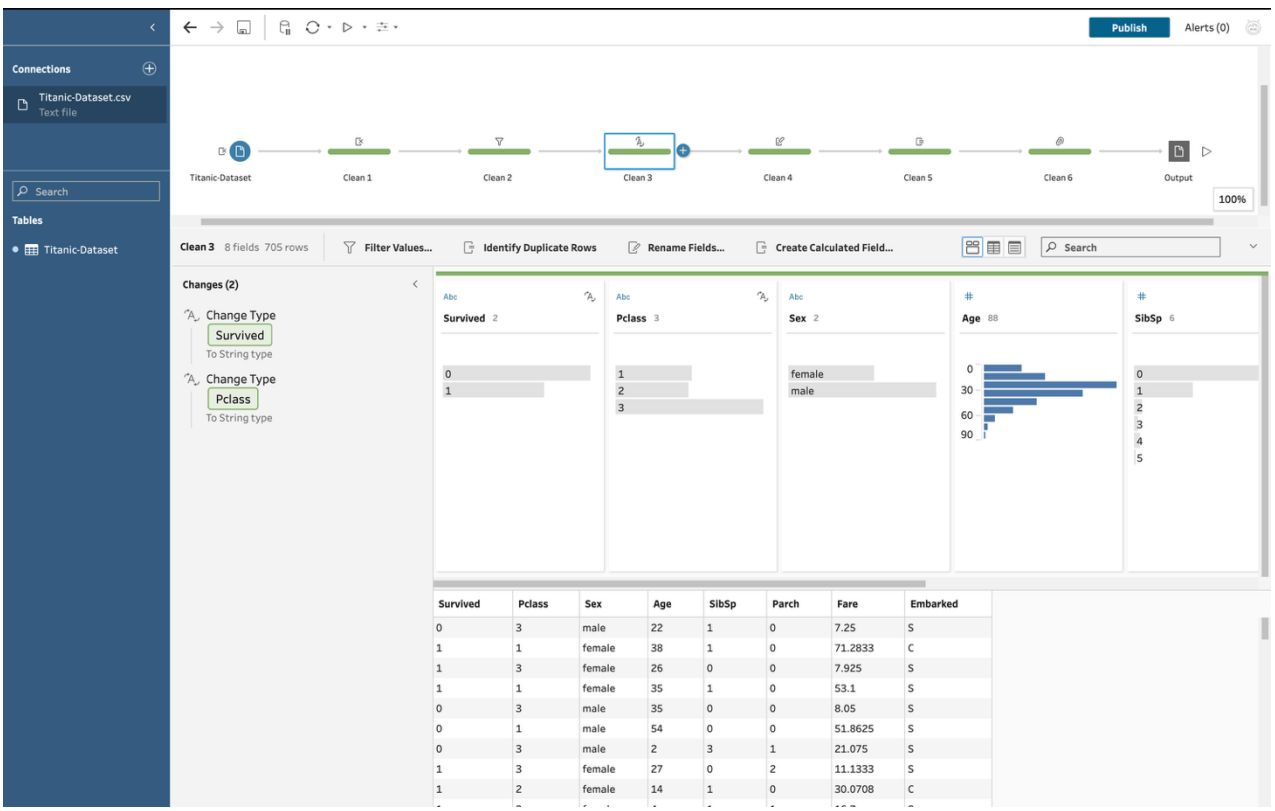
2. **Filter – Fare**: Condition [Fare] > 0 → removes rows where fare is zero (invalid/erroneous entries)
3. **Filter Null Values – Age**: Keep only non-null values → removes 177 rows with missing age

After this step, the dataset is reduced to **705 clean rows** with no null values in any remaining field.

Quality Dimension Improved: Completeness, Validity

Clean 3 – Fix Data Types (705 rows, 8 fields)

Changes (2): Change Type



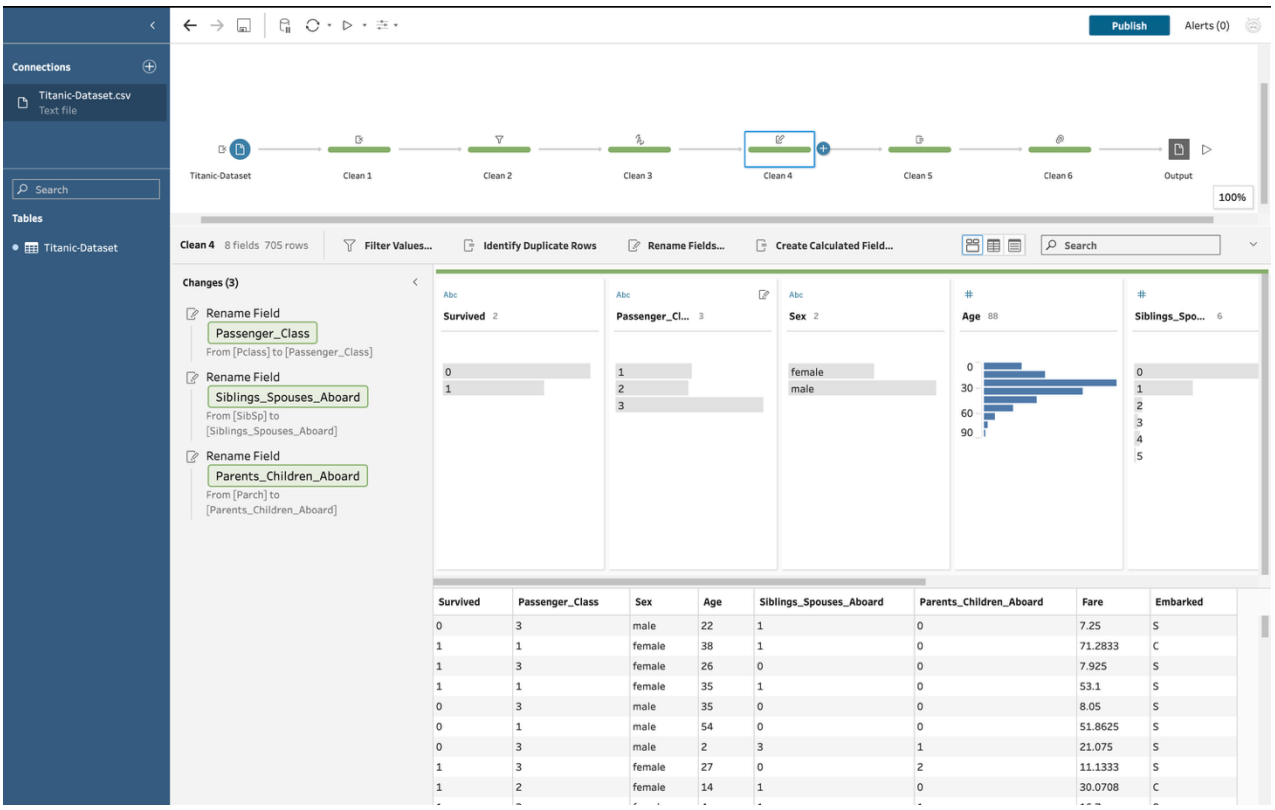
- Survived: Number → **String** (values 0/1 represent "No"/"Yes" — categorical, not numeric)
- Pclass: Number → **String** (values 1/2/3 represent passenger class categories, not quantities)

Both fields now display the Abc (String) type icon in the Profile pane, confirming the correction.

Quality Dimension Improved: Validity

Clean 4 – Rename Fields (705 rows, 8 fields)

Changes (3): Rename Field



All abbreviated or unclear field names are renamed for clarity:

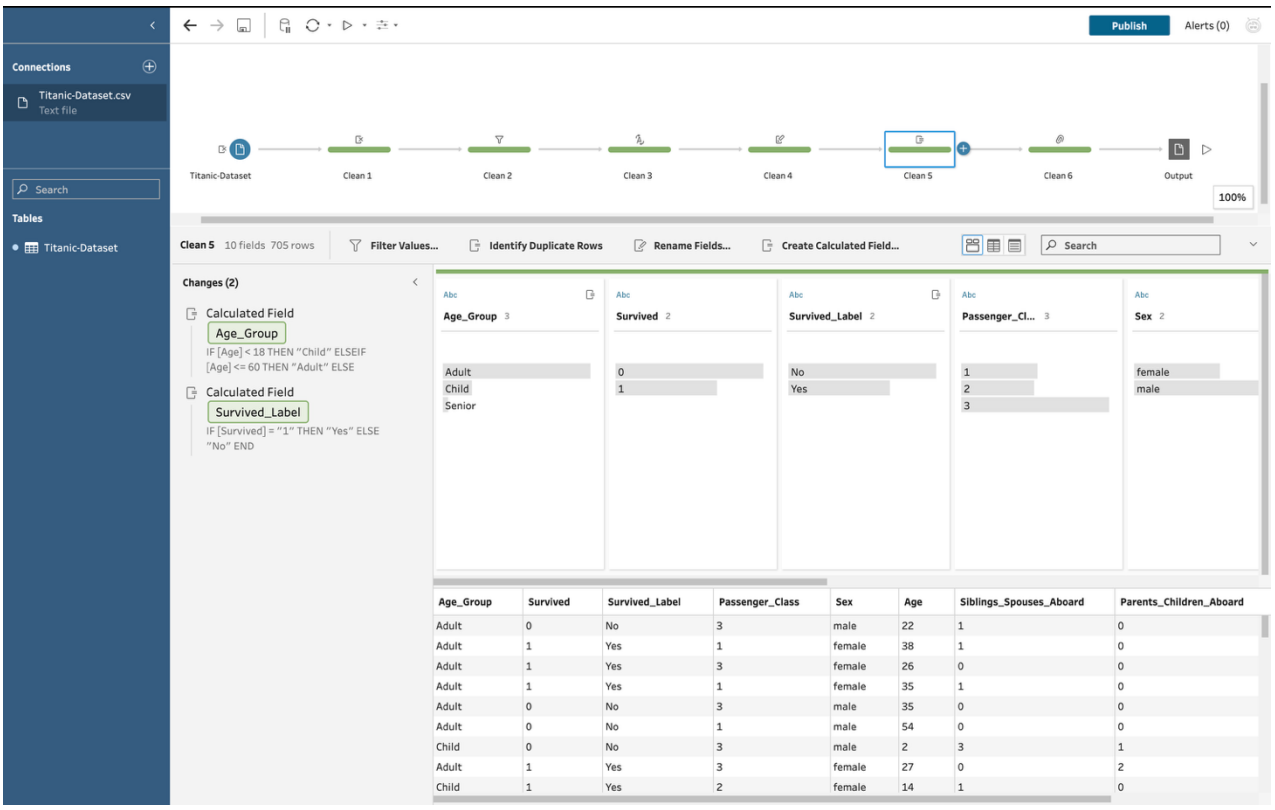
Original Name	Renamed To
Pclass	Passenger_Class
SibSp	Siblings_Spouses_Aboard
Parch	Parents_Children_Aboard

Tableau Prep automatically logs all renaming actions in the **Changes** panel, providing a traceable transformation history.

Quality Dimension Improved: Traceability, Consistency

Clean 5 – Create Calculated Fields (705 rows, 8 → 10 fields)

Changes (2): Calculated Field



Two new calculated fields are created using **Create Calculated Field**:

Field 1 – Age_Group:

```
IF [Age] < 18 THEN "Child"
ELSEIF [Age] <= 60 THEN "Adult"
ELSE "Senior"
END
```

Bins the continuous Age field into three meaningful categories: Child, Adult, Senior.

Field 2 – Survived_Label:

```
IF [Survived] = "1" THEN "Yes"
ELSE "No"
END
```

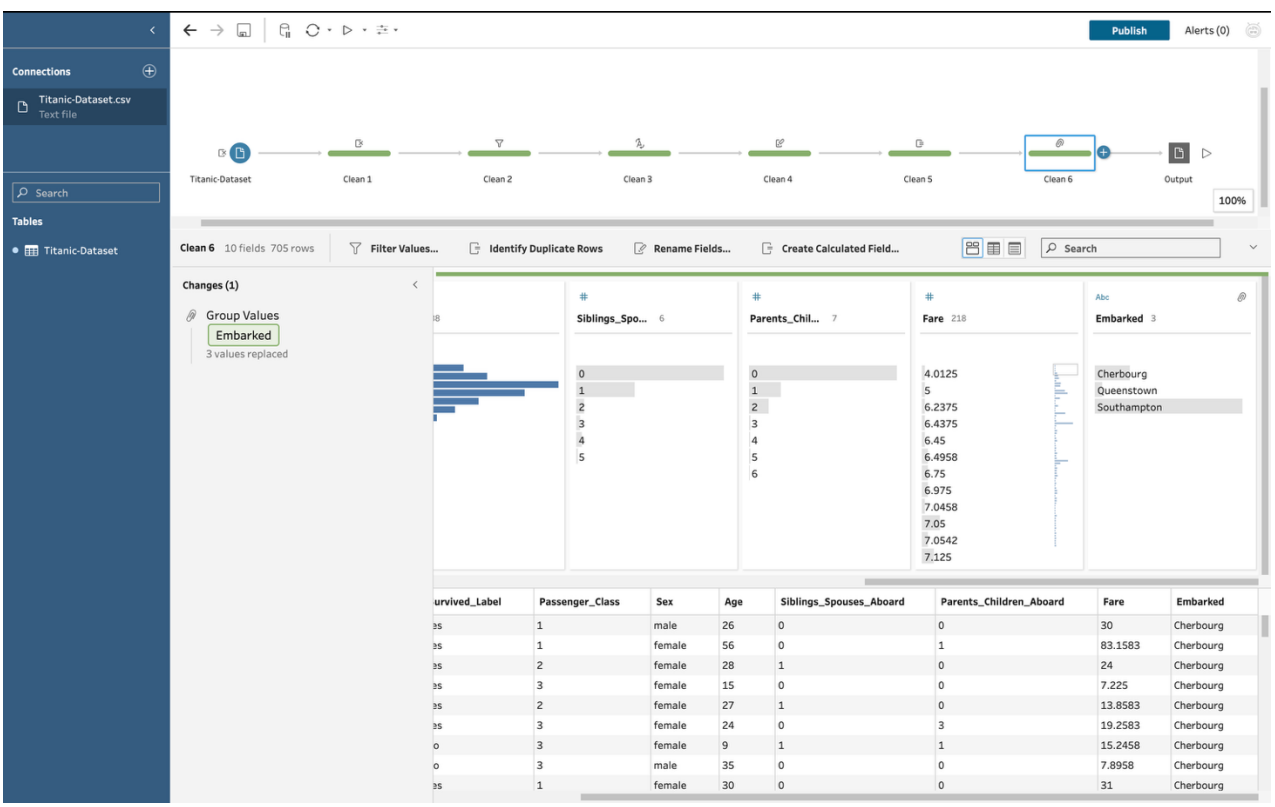
Converts the binary Survived field into human-readable labels ("Yes" / "No").

After this step, the dataset has **10 fields** and the Profile pane confirms three distinct Age_Group values and two Survived_Label values.

Quality Dimension Improved: Validity, Consistency, Traceability

Clean 6 – Standardise Embarked Values (705 rows, 10 fields)

Changes (1): Group Values – Embarked



The abbreviated port codes in the Embarked field are replaced with full descriptive names using **Group and Replace**:

Original Value	Replaced With
C	Cherbourg
Q	Queenstown
S	Southampton

A total of **3 values replaced**. The Profile pane now displays the full port names, making the data self-explanatory without requiring a separate data dictionary.

Quality Dimension Improved: Consistency, Validity

Output – Export as CSV

An Output step is added at the end of the flow. The output format is set to **CSV** and the file is saved as output.csv. The flow is executed by clicking **Run Flow**.

Final dataset: 705 rows, 10 fields

4. Data Quality Dimensions: Summary of Coverage

Dimension	How It Was Addressed
Completeness	Removed Cabin (77% null); filtered null Embarked rows (2); filtered null Age rows (177)
Validity	Corrected data types for Survived and Pclass; filtered zero Fare values; created Age_Group and Survived_Label calculated fields
Relevance	Removed PassengerId, Name, Ticket, Cabin — no analytical purpose

Consistency	Standardised all field names; replaced Embarked abbreviations (C/Q/S) with full port names
Traceability	Renamed fields to self-explanatory labels; all 15 transformations logged automatically in Tableau Prep's Changes panel
Uniqueness	Verified no duplicate rows exist via the Profile pane row count inspection

5. Conclusion

Main Problems Found

The Titanic dataset contained eight distinct data quality issues across completeness, validity, relevance, consistency, and traceability dimensions. The most critical issues were the Cabin field (77% null), 177 missing Age values, incorrect integer data types for categorical fields (Survived, Pclass), abbreviated port codes in Embarked, and four irrelevant identifier/text fields.

Steps Applied

A 6-step Tableau Prep Builder flow was constructed:

1. **Clean 1** – Removed 4 irrelevant/sparse fields (PassengerId, Name, Cabin, Ticket)
2. **Clean 2** – Filtered out null Embarked rows, zero Fare rows, and null Age rows → reduced to 705 rows
3. **Clean 3** – Corrected data types for Survived and Pclass (Number → String)
4. **Clean 4** – Renamed 3 abbreviated fields to descriptive names
5. **Clean 5** – Created 2 calculated fields: Age_Group and Survived_Label
6. **Clean 6** – Replaced 3 Embarked abbreviations with full port names

Improvements Achieved

- Dataset reduced from 12 to 10 meaningful, well-named fields

- Row count reduced from 891 to **705 fully complete rows** with no null values in any field
- All fields carry correct and appropriate data types
- Port names are now human-readable (Cherbourg, Queenstown, Southampton)
- Two new calculated fields add structured analytical value (Age_Group, Survived_Label)
- All 15 transformation steps are logged and traceable in the Changes panel

Remaining Limitations

Filtering the 177 null Age rows reduced the dataset by approximately 20% (from 891 to 705 rows), which may introduce a slight selection bias since the missing age values were not randomly distributed. In a production pipeline, mean or median imputation would be preferred, but Tableau Prep's basic toolset does not support this directly. Additionally, extreme high-fare outliers (e.g., £512) were retained in the dataset, as outlier removal was beyond the scope of the in-class techniques.

Report prepared as part of the Data Quality Management course assignment. Dataset sourced from Kaggle (https://www.kaggle.com/datasets/yasserh/titanic-dataset). All workflow steps performed in Tableau Prep Builder.